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Aufgabe 1.1

Let $N = (S, T, W)$ be a Petri net with connectivity matrix $\mathbb{C}$ and $M_1, M_2 \in \mathbb{N}^{|S|}, \sigma = t_i \dots t_m \in T^*$ such that $M_1[\sigma\rangle M_2$. To prove: $M_2 = M_1 + \mathbb{C} \cdot \psi(\sigma)$

Proof. From the definition of the connectivity matrix it follows that:

$$\mathbb{C} = \Big(W(t_1, -) - W(-, t_1) \dots W(t_{|T|}, -) - W(-, t_{|T|})\Big)$$

Thus the term $\mathbb{C} \cdot \psi(\sigma)$ results in

$$\mathbb{C} \cdot \psi(\sigma) = (W(t_i, -) - W(-, t_i) + \dots + W(t_m, -) - W(-, t_m))$$

and therefore according to the definition of the firing relation:

$$M_1[\sigma\rangle M_2 = M_1 + W(t_i, -) - W(-, t_i) + \dots + W(t_m, -) - W(-, t_m) = M_1 + \mathbb{C} \cdot \psi(\sigma)$$

Thus the proposition is proven. □

Aufgabe 1.2

Aufgabe 1.3

Aufgabe 1.4